The Fractional Laplacian: Rigorous Construction and Physical Origin of the Half-Order Diffusion Operator

Supplementary Annex

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Abstract

Section 2 of the main text reduces the governing equation for inter-rifting deformation to $\partial_t \hat{u} = -a|k|\hat{u}$ in Fourier space, and identifies $|k|$ as the symbol of the half-Laplacian $(-\Delta)^{1/2}$. This annex grounds that identification rigorously, following the definitions and equivalences surveyed by Daoud and Laamri (2022). The central result is that the operator order $s = 1/2$ is not a modeling choice: the Caffarelli–Silvestre extension (Caffarelli and Silvestre, 2007) that represents $(-\Delta)^s$ as a Dirichlet-to-Neumann map degenerates, exactly at $s = 1/2$, to the classical unweighted harmonic extension problem, the same half-space potential problem underlying the Newtonian viscous half-space coupling of Savage and Prescott (1978). We give short proofs of the results that are load-bearing for the model's own construction, cite the rest, and synthesize the mathematics for a structural-geology reading of the model.

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1 The kernel is the fractional Laplacian

1.1 Recap of the operator

Section 2.1–2.2 of the main text derives the governing equation for the surface displacement $u(x,t)$ of an elastic plate over a Newtonian viscous half-space, and reduces it in Fourier space to

$$\frac{\partial \hat{u}}{\partial t} = -a|k|\hat{u}, \tag{1.1}$$

with $a = EH/[2\eta(1-\nu^2)]$. The wavenumber symbol $|k|$ is identified there, by reference to the classical fractional-diffusion literature (Metzler and Klafter, 2000; Samko et al., 1993), as the Fourier symbol of the half-Laplacian $(-\Delta)^{1/2}$. This annex does not re-derive the physical setup; it establishes, from the definitions surveyed by Daoud and Laamri (2022), which operator this is and why its order is forced to be $s = \frac{1}{2}$ by the coupling's own geometry, not a relabeling of an already-known kernel.

1.2 The normalizing constant at $N = 1$, $s = \frac{1}{2}$

Daoud and Laamri (2022) give the fractional Laplacian on $\mathbb{R}^N$ as the hypersingular principal-value integral. Here and for the rest of this annex, v denotes a generic test function on which the operator acts, following Daoud and Laamri's own notation; it is unrelated to the surface velocity $\partial u/\partial t$ or to the main text's v^{STEP}, which never appears in this abstract role.

$$(-\Delta)^s v(x) = C(N,s)\ \mathrm{P.V.}\int_{\mathbb{R}^N} \frac{v(x)-v(y)}{\|x-y\|^{N+2s}}\,dy, \qquad C(N,s) = \frac{s\,4^s}{\pi^{N/2}}\,\frac{\Gamma(s+\frac{N}{2})}{\Gamma(1-s)}. \tag{1.2}$$

At $N = 1$, $s = \frac{1}{2}$,

$$C\big(1,\tfrac{1}{2}\big) = \frac{\frac{1}{2}\cdot 4^{1/2}}{\pi^{1/2}}\cdot\frac{\Gamma(1)}{\Gamma(\frac{1}{2})} = \frac{1}{\sqrt{\pi}}\cdot\frac{1}{\sqrt{\pi}} = \boxed{\frac{1}{\pi}}, \tag{1.3}$$

using $\Gamma(1) = 1$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Confirmed numerically in `verify_fractional_kernel.jl`. So, for $N = 1$ and $s = \frac{1}{2}$, (1.2) reads

$$(-\Delta)^{1/2} v(x) = \frac{1}{\pi}\ \mathrm{P.V.}\int_{-\infty}^{\infty} \frac{v(x)-v(y)}{(x-y)^2}\,dy. \tag{1.4}$$

The factor of π in the denominator of Savage's kernel (main text eq. 3, $2\eta/[\pi(x-x')^2]$) is not a rheological constant: it is exactly the normalizing constant that makes the singular integral definition of the half-Laplacian equal its Fourier-multiplier definition.

1.3 Reconciling the two principal values

There is a subtlety that must be handled carefully rather than glossed over. The Stokes kernel convolution (main text eq. 3) is

$$\sigma_{xz}^{\mathrm{BOT}}(x,t) = \int_{-\infty}^{\infty} \frac{2\eta}{\pi(x-x')^2}\,\frac{\partial u}{\partial t}(x',t)\,dx', \tag{1.5}$$

described, following Savage and Prescott (1978), as a Cauchy principal value acting directly on $\partial u/\partial t$. Definition (1.4) instead takes the principal value of the *difference* $v(x)-v(y)$, not of $v(y)$ alone. These are not the same regularization, and the difference is not cosmetic: a genuine symmetric-truncation principal value of $\int v(y)/(x-y)^2\,dy$ diverges for any $v(x) \neq 0$, because $1/(x-y)^2$ is positive on both sides of the singularity and the truncated integral grows as $2v(x)/\varepsilon$ with no cancellation. What (1.5) calls a principal value is, precisely, the Hadamard finite part,

$$\mathrm{Hfp}\int_{-\infty}^{\infty} \frac{v(y)}{(x-y)^2}\,dy := \lim_{\varepsilon\to 0^+}\left[\int_{|y-x|\geq\varepsilon} \frac{v(y)}{(x-y)^2}\,dy - \frac{2v(x)}{\varepsilon}\right], \tag{1.6}$$

a regularization standard in dislocation and crack theory for this order-2 kernel, loosely called a "principal value" in that same literature.

The two regularizations are related by a clean sign identity. Expanding the principal value of the difference form,

$$\begin{aligned}\text{P.V.}\int \frac{v(x)-v(y)}{(x-y)^2}\,dy &= \lim_{\varepsilon\to 0^+}\left[v(x)\int_{|y-x|\geq\varepsilon}\frac{dy}{(x-y)^2}-\int_{|y-x|\geq\varepsilon}\frac{v(y)}{(x-y)^2}\,dy\right]\\ &= \lim_{\varepsilon\to 0^+}\left[\frac{2v(x)}{\varepsilon}-\int_{|y-x|\geq\varepsilon}\frac{v(y)}{(x-y)^2}\,dy\right] = -\,\text{Hfp}\int\frac{v(y)}{(x-y)^2}\,dy,\end{aligned}\tag{1.7}$$

using $\int_{|z|\geq\varepsilon} dz/z^2 = 2/\varepsilon$ exactly, for every $\varepsilon > 0$. Confirmed numerically in `verify_fractional_kernel.jl` for a Gaussian test function. Combining (1.4) and (1.7),

$$\text{Hfp}\int\frac{v(y)}{(x-y)^2}\,dy = -\pi\,(-\Delta)^{1/2}v(x).\tag{1.8}$$

Applying (1.8) to $v = \partial u/\partial t$ in the Stokes kernel (1.5),

$$\sigma_{xz}^{\text{BOT}}(x,t) = \frac{2\eta}{\pi}\,\text{Hfp}\int\frac{\partial u/\partial t\,(x')}{(x-x')^2}\,dx' = \frac{2\eta}{\pi}\left[-\pi\,(-\Delta)^{1/2}\frac{\partial u}{\partial t}\right] = -2\eta\,(-\Delta)^{1/2}\frac{\partial u}{\partial t}.\tag{1.9}$$

In Fourier space, $(-\Delta)^{1/2}$ has symbol $|k|$, so $\mathcal{F}\{\sigma_{xz}^{\text{BOT}}\} = -2\eta|k|\,\partial\hat{u}/\partial t$, the classical result quoted in the main text (eq. 5) and in Savage and Prescott (1978). The Stokes kernel is therefore -2η times the half-Laplacian of the surface velocity field, term for term and with the correct sign. This is not a loose spectral analogy: identifying the regularization used in the geophysics literature with the Hadamard finite part implicit in the singular-integral definition of $(-\Delta)^{1/2}$ shows the two are the same operator.

This identity holds for the Stokes term alone, and it is naturally a statement about velocity rather than displacement: a viscous material resists the *rate* of deformation, not the deformation itself, so the basal traction depends on $\partial u/\partial t$, unlike the elastic term (main text eq. 1, Hooke's law), which depends on u directly. Combining the two in the full governing equation (main text eq. 4) and Fourier-transforming is what closes the loop back to (1.1). Because $(-\Delta)^{1/2}$ is a purely spatial, time-independent operator, it commutes with $\partial/\partial t$: $(-\Delta)^{1/2}(\partial u/\partial t) = \partial_t\big[(-\Delta)^{1/2}u\big]$. The combined equation therefore rearranges into an equation purely in u, $\partial_t\hat{u} = -a|k|\hat{u}$: the same half-Laplacian that maps velocity to traction in the viscous term also governs the time evolution of displacement itself, once the elastic restoring term is folded in. From here on, this annex works with that displacement-evolution form directly (the initial-value problem for $u(x,t)$ given $u(x,0)$), since that is the equation whose solution is the object of physical interest; the velocity field never needs to reappear as a separate unknown.

2 The correct function space: $H^{1/2}(\mathbb{R})$

Daoud and Laamri (2022) identify $H^s(\mathbb{R}^N)$, equipped with the Gagliardo seminorm

$$[v]^2_{H^s(\mathbb{R}^N)} := \int_{\mathbb{R}^N}\int_{\mathbb{R}^N}\frac{|v(x)-v(y)|^2}{\|x-y\|^{N+2s}}\,dx\,dy,\tag{2.1}$$

as the natural energy space for $(-\Delta)^s$: their Proposition 2.6 relates the seminorm directly to the operator, $[v]^2_{H^s} = 2\,C(N,s)^{-1}\|(-\Delta)^{s/2}v\|^2_{L^2}$ (see also Di Nezza et al., 2012 for the full development of these spaces). For the model's own construction, $N = 1$, $s = \frac{1}{2}$, so the relevant space is $H^{1/2}(\mathbb{R})$,

$$[v]^2_{H^{1/2}(\mathbb{R})} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\frac{|v(x)-v(y)|^2}{(x-y)^2}\,dx\,dy.\tag{2.2}$$

2.1 The dike-opening initial condition is not in $H^{1/2}(\mathbb{R})$

The initial condition for a single dike intrusion is a pure step. Following the main text's own convention (eq. 11), $u(x,0^+) = b\,H(-x)$, where $H(-x)$ is the Heaviside function (1 for $x < 0$, 0 for $x > 0$); we write it as plain H throughout this annex, reserving the calligraphic $\mathcal{H}$ for the Hilbert transform introduced in Section 4 so the two are never confused.

The Gagliardo seminorm is quadratic in its argument: for any constant c and function w, $(cw)(x) - (cw)(y) = c\,(w(x) - w(y))$, so c^2 factors out of the double integral in (2.2) unchanged and $[cw]^2_{H^{1/2}(\mathbb{R})} = c^2\,[w]^2_{H^{1/2}(\mathbb{R})}$. It is therefore enough to compute the seminorm of the unit-amplitude step $v := H(-\cdot)$ (the function $x \mapsto H(-x)$, with $b = 1$) and rescale by b^2 at the end, rather than carrying the physical amplitude b through every line of the calculation. Substituting $v = H(-\cdot)$ into (2.2): for $x > 0$, $y < 0$ (or vice versa), $v(x) = 0$ and $v(y) = 1$, so $|v(x) - v(y)|^2 = 1$; for x, y of the same sign, the integrand vanishes. So

$$[H(-\cdot)]^2_{H^{1/2}(\mathbb{R})} = 2\int_0^\infty \int_0^\infty \frac{dt\,dx}{(x+t)^2}, \qquad y = -t,\ t > 0, \tag{2.3}$$

using the symmetry of the integrand under $(x, y) \mapsto (y, x)$ to double the $x > 0, y < 0$ contribution. The inner integral is elementary and exact for every $x > 0$:

$$\int_0^\infty \frac{dt}{(x+t)^2} = \frac{1}{x}. \tag{2.4}$$

The outer integral $\int_0^\infty dx/x$ diverges logarithmically as $x \to 0^+$ (confirmed numerically in `verify_sobolev_divergence.jl`: the truncated integral $I(\delta) = \int_\delta^\infty dx/x$ tracks $-\log\delta$ to within a fixed offset across four orders of magnitude in δ). Since $u(x, 0^+) = b\,H(-x)$ differs from $H(-\cdot)$ only by the constant scale factor b,

$$[u(\cdot, 0^+)]^2_{H^{1/2}(\mathbb{R})} = b^2\,[H(-\cdot)]^2_{H^{1/2}(\mathbb{R})} = +\infty. \tag{2.5}$$

The pure step is the textbook borderline case: $H^{1/2}(\mathbb{R})$ is the critical exponent in one dimension, the threshold at which the fractional Sobolev embedding fails to control jump discontinuities by only a logarithm (Di Nezza et al., 2012). This is not a defect of the model. It is a precise statement of a fact the model already handles correctly by construction: the solution is not required to remain in the natural energy space at $t = 0$, because the evolution operator smooths it instantaneously for any $t > 0$, which we establish rigorously in Section 3.

This raises a natural question: if $u(\cdot, 0^+) \notin H^{1/2}(\mathbb{R})$, where does it actually live, and does the failure matter? The two parts of the question have different answers depending on whether "living" is asked locally or globally. Locally, near the jump alone, the fractional Sobolev embedding fails only at the critical exponent: repeating the computation above with a generic exponent s in place of $\frac{1}{2}$, the inner integral becomes $\int_0^\infty dt/(x+t)^{1+2s} = x^{-2s}/(2s)$, whose outer integral over x near 0 converges for every $s < \frac{1}{2}$ and diverges (worse than logarithmically) for every $s > \frac{1}{2}$; $s = \frac{1}{2}$ is the one exponent at which it diverges only logarithmically. So $u(\cdot, 0^+) \in H^s_{\mathrm{loc}}(\mathbb{R})$ for every $s < \frac{1}{2}$, and fails only at, and above, the critical exponent. Globally, the more mundane fact that $H(-x)$ never decays as $x \to \pm\infty$ keeps it out of $L^2(\mathbb{R})$, and hence out of $H^s(\mathbb{R})$ for any s, for a reason that has nothing to do with the jump; the natural global home for a bounded function with a single finite jump and no other structure is the space of bounded variation, $BV(\mathbb{R})$. So,

$$u(\cdot, 0^+) \in BV(\mathbb{R}) \cap \bigcap_{s<1/2} H^s_{\mathrm{loc}}(\mathbb{R}), \qquad u(\cdot, 0^+) \notin H^{1/2}(\mathbb{R}),$$

the classical statement that $BV(\mathbb{R})$ fails to embed into the critical Sobolev space $H^{1/2}(\mathbb{R})$ by exactly a logarithm (Di Nezza et al., 2012).

Whether this matters depends on which route is used to solve the equation. It does not matter for the route this annex takes: the semigroup formula of Section 3 is defined for any tempered distribution as initial data, $BV(\mathbb{R})$ included, with no need for the initial condition to sit in the equation's own energy space, and the semigroup's smoothing repairs whatever regularity the initial data lacks the instant t becomes positive. It would matter for a different route: classical energy or variational methods for parabolic-type equations (Galerkin approximation, weak formulations built directly on the natural energy space) generally require the initial data to already sit in that energy space to set up the weak problem cleanly, and a rigorous treatment via that route would need to work around this initial condition explicitly, for instance by posing the problem in $BV(\mathbb{R})$ directly or by falling back on the semigroup formula used here.

3 Construction via the semigroup

This section proceeds in four steps. Section 3.1 constructs the model's solution directly and identifies it as a genuine semigroup. Section 3.2 gives an equivalent, second construction that puts the heat-semigroup function w of (3.3) to explicit use, rather than leaving it as an unused ingredient of a definition. Section 3.3 shows what the semigroup does at high spatial frequency: it smooths, with an explicit, growing radius of analyticity. Section 3.4 shows what it does not touch, at zero frequency, and why that is harmless.

3.1 Construction of the solution

Equation (1.1) is not a postulate about an abstract operator: it is the model's own momentum balance, already established in Section 1, Hooke's law for the elastic plate combined with the Stokes-kernel coupling to the viscous half-space, Fourier-transformed. Solving it needs nothing beyond ordinary linear ODE theory. For each fixed wavenumber k, $\partial_t \hat{u} = -a|k|\hat{u}$ integrates directly to

$$\hat{u}(k,t) = \hat{u}(k,0)\, e^{-a|k|t}. \tag{3.1}$$

This is the model's solution, in full; nothing that follows in this section adds to it. The family of multipliers $S(t) : k \mapsto e^{-a|k|t}$, $t \geq 0$, satisfies the semigroup law exactly,

$$S(t_1)S(t_2) = e^{-a|k|t_1} e^{-a|k|t_2} = e^{-a|k|(t_1+t_2)} = S(t_1+t_2), \qquad S(0) = 1, \tag{3.2}$$

identifying it as the semigroup generated by $-a(-\Delta)^{1/2}$: solving the physical equation and identifying the operator it contains are two separate facts, and (3.2) records the second, not a further step in the first.

That the operator appearing in (1.1) is the same $(-\Delta)^{1/2}$ characterized throughout this annex, rather than merely something that behaves like it in Fourier space, is what Section 1 already proves directly for this specific kernel. A further, independent confirmation comes from the general theory of the operator itself, unrelated to the model or to u. Daoud and Laamri (2022) give a definition of $(-\Delta)^s$, for any generic function v, through the heat semigroup: for $w(x,\tau)$ solving the classical heat equation $\partial_\tau w = \partial_x^2 w$ with $w(x,0) = v(x)$,

$$(-\Delta)^s v(x) = \frac{s}{\Gamma(1-s)} \int_0^\infty \big(v(x) - w(x,\tau)\big)\, \frac{d\tau}{\tau^{1+s}}. \tag{3.3}$$

Here τ is an internal, dummy diffusion time used only to define the operator at a single instant; it has no relation to the model's physical evolution time t, and v is any test function, not the model's own u. Equation (3.3) is pure operator theory: it holds regardless of whether $(-\Delta)^{1/2}$ ever appears in a physical equation, and it plays no role in solving (1.1), which was already solved above with no reference to w at all. Its purpose here is narrower and different: to complete the catalogue of equivalent constructions of $(-\Delta)^{1/2}$ (alongside the singular integral of Section 1 and the Fourier multiplier), and to set up a second, genuinely useful role for w in the model's own solution, taken up next.

3.2 An equivalent view: Bochner subordination and the role of w

There is a second, equally valid route to (3.1) that puts w to direct use, rather than leaving it stranded in (3.3). Bochner's subordination formula writes the semigroup generated by $-a(-\Delta)^{1/2}$ as a weighted average, over all internal diffusion times τ, of the ordinary heat semigroup:

$$u(x,t) = \int_0^\infty w(x,\tau)\, f_{at}(\tau)\, d\tau, \qquad f_{at}(\tau) = \frac{at}{2\sqrt{\pi}}\, \tau^{-3/2}\, e^{-(at)^2/(4\tau)}, \tag{3.4}$$

where $w(x,\tau)$ is the ordinary (Fickian) heat-equation solution with the model's own initial condition, $w(x,0) = u(x,0^+) = b\,H(-x)$, and f_{at} is the classical density of the $\frac{1}{2}$-stable subordinator at "time" at: the same w as in (3.3), now integrated against an explicit weight instead of differenced against a single value of v, so that (3.3) and (3.4) are the differential and integrated statements of the same subordination principle.

For the model's own initial condition, w has the classical closed form

$$w(x,\tau) = \frac{b}{2}\,\mathrm{erfc}\!\left(\frac{x}{2\sqrt{\tau}}\right), \tag{3.5}$$

obtained by convolving $u(x,0^+) = b\,H(-x)$ with the ordinary heat kernel; this is the Foulger diffusion profile the main text rejects as the physical model (§4.3, erfc tails rather than Lorentzian). It reappears here purely as a mathematical ingredient: the correct, non-Fickian solution $u(x,t)$ is not $w(x,t)$ itself, but the average of $w(x,\cdot)$ over all diffusion times τ, weighted by the subordinator density f_{at}.

Two independent checks confirm (3.4), at two different levels. The first is general, at the level of the operator itself, and does not reference the model's specific initial condition. The ordinary heat semigroup has Fourier multiplier $e^{-k^2\tau}$ (a direct consequence of $\partial_\tau w = \partial_x^2 w$); averaging it against $f_{at}(\tau)$ is exactly a Laplace transform of f_{at} in τ, evaluated at $p = k^2$:

$$\int_0^\infty e^{-k^2\tau} f_{at}(\tau)\,d\tau = e^{-at|k|}, \tag{3.6}$$

confirmed to machine precision for $k = 0.3, 1, 2.5, 5$. The right-hand side is exactly the multiplier of (3.1), so this identity alone already guarantees that (3.4) and (3.1) agree for *any* initial data v, not only $u(x,0^+)$.

The second check applies that general fact to the model's own concrete quantities. It is not logically required once (3.6) is established, but it ties the abstract operator identity back to the specific numbers the model uses. Substituting the closed form (3.5) into (3.4) gives the explicit, one-dimensional integral

$$u(x,t) = \int_0^\infty \frac{b}{2}\,\mathrm{erfc}\!\left(\frac{x}{2\sqrt{\tau}}\right) \frac{at}{2\sqrt{\pi}}\,\tau^{-3/2}\,e^{-(at)^2/(4\tau)}\,d\tau, \tag{3.7}$$

with no unresolved symbols left: every quantity on the right is a known function of τ for fixed x, a, b, t. Evaluating (3.7) by numerical quadrature reproduces the main text's own closed form $u(x,t) = \frac{b}{2} - \frac{b}{\pi}\arctan(x/(at))$ to 10^{-9} across $x \in [-3,3]$ (quadrature tolerance).

The subordination view is not needed for anything that follows, but it is what makes the smoothing established next intuitive rather than merely algebraic, since each $w(x,\tau)$, $\tau > 0$, is already smooth by ordinary heat-kernel diffusion, and an average of smooth functions is smooth.

3.3 High-frequency damping and analyticity

Return to the direct construction (3.1). For fixed $t > 0$ and any integer $n \geq 0$, maximizing $k^n e^{-akt}$ over $k > 0$ gives the sharp bound

$$\sup_{k\in\mathbb{R}} |k|^n e^{-a|k|t} = \left(\frac{n}{ate}\right)^n < \infty, \tag{3.8}$$

confirmed numerically to 10^{-10} precision for $n = 1, 2, 3, 5$. Because this supremum is finite for every n, $|k|^n e^{-a|k|t} \to 0$ as $|k| \to \infty$ for every n: the multiplier decays faster than any power of $|k|$, uniformly in k.

This decay controls the local smoothness of $u(\cdot,t)$ directly, and makes it precise enough to quote a radius, not just an adjective. Extending $x \mapsto x + i\sigma$ in (3.1) and inverting,

$$u(x+i\sigma,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{u}(k,0)\,e^{-a|k|t-k\sigma}\,e^{ikx}\,dk, \tag{3.9}$$

the exponent $-a|k|t - k\sigma$ tends to $-\infty$ as $|k| \to \infty$ exactly when $|\sigma| < at$: it grows as $-k(at+\sigma)$ for $k \to +\infty$ and as $-|k|(at-\sigma)$ for $k \to -\infty$, both negative if and only if $-at < \sigma < at$. So (3.9) converges, and $u(\cdot,t)$ extends holomorphically, throughout the strip

$$|\mathrm{Im}\,z| < at, \tag{3.10}$$

confirmed numerically: truncating (3.9) at $|k| \leq K$, the truncated integral converges to a finite limit as $K \to \infty$ for every $|\sigma| < at$ (e.g. stabilizing near 10.5 for $\sigma/at = 0.9$), grows linearly in K exactly at $\sigma = at$, and diverges super-exponentially for $|\sigma| > at$ (already 10^{43} at $K = 200$ for $\sigma/at = 1.5$). A

function holomorphic throughout a strip around the real axis is, in particular, real analytic on the real axis; (3.10) makes this quantitative: the radius of convergence of $u(\cdot,t)$'s Taylor series at any point $x \in \mathbb{R}$ is at least at, growing linearly in time, no matter how singular $u(\cdot,0)$ is. The dike intrusion initial condition $u(x,0^+) = b\,H(-x)$ has a jump at $x = 0$ and $t = 0$, and that discontinuity is erased completely (radius of convergence $at > 0$ everywhere) the instant t becomes positive. This resolves the tension flagged in Section 2.1: $u(x,0^+) = b\,H(-x)$ fails to belong to the natural energy space $H^{1/2}(\mathbb{R})$, yet $u(\cdot,t)$ is real analytic in x for every $t > 0$. The same conclusion applies to the closed-form step response given in the main text (§2.3): its printed formula is a single-branch simplification valid for $x > 0$, but the solution it represents, correctly extended to all x, is the real analytic function (3.10) predicts.

3.4 What survives at $k = 0$

Everything in Section 3.3 concerned the limit $|k| \to \infty$; the opposite limit, $k \to 0$, is where the multiplier $e^{-a|k|t}$ is least effective; it equals 1 there for every t, so whatever singularity $\hat{u}(k,0)$ carries at $k = 0$ is carried forward unchanged, for all time. This is not a gap in the argument above: it is a separate question, about a different part of the spectrum, with its own separate (and equally explicit) answer. The main text's own Fourier transform of the initial condition (eq. 12) is $\hat{u}(k,0) = b\pi\delta(k) + ib/k$, and $e^{-a|k|t} = 1$ at $k = 0$ for every t, so neither term is ever damped there. The Dirac term $b\pi\delta(k)$ alone is enough to show $\hat{u}(\cdot,t)$ is never Schwartz class, for any $t \geq 0$: a genuine delta distribution is not even a locally bounded function, let alone rapidly decaying. It corresponds, on the spatial side, to the constant background $b/2$ appearing in the inverse transform of eq. 13's delta term (main text §2.3), which is trivially smooth (a constant has infinite radius of convergence everywhere) and carries no discontinuity. The pole ib/k plays a complementary, quantitative role. Using the main text's globally correct closed form $u(x,t) = b/2 - (b/\pi)\arctan(x/(at))$ (the analytic continuation of eq. 15–16 to all x) and the large-argument expansion $\arctan(z) = \pi/2 - z^{-1} + O(z^{-3})$,

$$u(x,t) - \frac{b}{2} \;\sim\; \frac{abt}{\pi x} \qquad (x \to +\infty,\ t \text{ fixed}), \tag{3.11}$$

confirmed numerically: the ratio of the two sides is 0.996 at $x = 10$, 0.999996 at $x = 100$, and $1 - 4\times10^{-7}$ at $x = 1000$ (for $a = 1.3, b = 1, t = 0.8$). The algebraic, $1/x$ decay in (3.11) is the spatial signature of a simple pole at $k = 0$: low-frequency singularities in the spectrum correspond to algebraic, not exponential, far-field tails. Neither survivor at $k = 0$, the delta or the pole, threatens the analyticity bound (3.10), which is controlled entirely by the decay of the multiplier at $|k| \to \infty$ established in (3.8) and is completely insensitive to whatever happens at $k = 0$. The semigroup construction shows this smoothing is not a special feature of one particular closed form: it is the generic behaviour of $(-\Delta)^{1/2}$-semigroups, quantified by (3.8)–(3.10) for any initial data, independent of whatever singularity the spectrum retains at $k = 0$.

4 Centerpiece: the Caffarelli–Silvestre extension

4.1 The half-Laplacian is the derivative of the Hilbert transform

Define the Hilbert transform $\mathcal{H}v(x) := \frac{1}{\pi}\,\mathrm{P.V.}\int_{-\infty}^{\infty} \frac{v(y)}{x-y}\,dy$, whose Fourier symbol, under the convention $\hat{v}(k) = \int v(x)e^{-ikx}dx$ used throughout this annex and the main text, is $-i\,\mathrm{sgn}(k)$. Composing with differentiation, whose symbol is ik,

$$\mathcal{F}\left\{\frac{d}{dx}\big[\mathcal{H}v\big]\right\}(k) = (ik)\cdot(-i\,\mathrm{sgn}(k))\,\hat{v}(k) = k\,\mathrm{sgn}(k)\,\hat{v}(k) = |k|\,\hat{v}(k), \tag{4.1}$$

which is the symbol of $(-\Delta)^{1/2}$. So, as operators on Schwartz functions,

$$(-\Delta)^{1/2}v = \frac{d}{dx}\big[\mathcal{H}v\big]. \tag{4.2}$$

Confirmed numerically via FFT in `verify_hilbert_symbol.jl`. This identity is specific to one spatial dimension: it is what makes the half-Laplacian, uniquely among fractional powers, expressible through

a classical, purely local-looking construction (a derivative of a boundary value of a harmonic function), which is the content of the next subsection.

4.2 The extension problem at $s = \frac{1}{2}$

Daoud and Laamri (2022) (their fifth definition, following Caffarelli and Silvestre, 2007) realize $(-\Delta)^s$ as a Dirichlet-to-Neumann map. Let $U(x, y)$ solve

$$\operatorname{div}\big(y^\alpha \nabla U\big) = 0 \quad \text{in } \mathbb{R} \times (0, \infty), \qquad U(x, 0) = v(x), \qquad \alpha = 1 - 2s \in (-1, 1). \tag{4.3}$$

Then

$$(-\Delta)^s v(x) = -d_s \lim_{y \to 0^+} y^\alpha \, \partial_y U(x, y), \qquad d_s = \frac{2^{2s-1}\Gamma(s)}{\Gamma(1-s)}. \tag{4.4}$$

At $s = \frac{1}{2}$: $\alpha = 1 - 2s = 0$ and $d_{1/2} = 2^0\,\Gamma(\frac{1}{2})/\Gamma(\frac{1}{2}) = 1$. The weight y^α disappears entirely, and (4.3) collapses to the classical, unweighted Laplace equation,

$$\boxed{\Delta U = 0 \quad \text{in } \mathbb{R} \times (0, \infty), \qquad U(x, 0) = v(x)}, \qquad (-\Delta)^{1/2} v(x) = -\lim_{y \to 0^+} \partial_y U(x, y). \tag{4.5}$$

This is not a special case chosen for convenience: it is the *only* value of $s \in (0, 1)$ for which the extension problem is the ordinary harmonic-extension problem in the upper half-plane, with no singular or vanishing weight at $y = 0$.

For $N = 1$, U is the classical Poisson extension of v into the upper half-plane, explicitly

$$U(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{v(x')}{(x - x')^2 + y^2} \, dx', \tag{4.6}$$

recovering v as $y \to 0^+$ and solving $\Delta U = 0$ throughout $y > 0$; and its harmonic conjugate V (so that $U + iV$ is holomorphic in $y > 0$) has boundary trace equal to the Hilbert transform of v, $V(x, 0) = \mathcal{H}v(x)$, a classical fact of harmonic and complex analysis. The Cauchy–Riemann relations give $-\partial_y U = \partial_x V$, so at $y = 0$,

$$-\lim_{y \to 0^+} \partial_y U(x, y) = \frac{d}{dx}\big[V(x, 0)\big] = \frac{d}{dx}\big[\mathcal{H}v\big](x), \tag{4.7}$$

which is $(-\Delta)^{1/2} v(x)$ by (4.2), confirming (4.5) is internally consistent with the independent Fourier-symbol derivation of Section 4.1.

4.3 Physical reading

Equation (4.5) says that solving for $(-\Delta)^{1/2}$ at the surface of $\mathbb{R}$ is the same problem as solving Laplace's equation in the half-space $y > 0$ above it and reading off the normal derivative at the boundary. That is, structurally, the same type of problem Savage's Stokes kernel already comes from: a linear response at the surface of a half-space, generated by potential theory in the half-space beneath it. Two-dimensional half-space Green's functions, whether for an elastic solid (the Boussinesq–Flamant problem) or for Stokes flow (via the Goursat representation of the biharmonic stream function as a pair of harmonic functions, the standard reduction used throughout half-space viscous flow theory), are built from this kind of harmonic/conjugate-harmonic machinery. This is a structural argument about why half-space linear response problems in two dimensions generically produce Hilbert-transform-type, half-Laplacian operators; it is not a claim to have reproduced Savage and Prescott's (1978) original derivation step by step. What Section 1 establishes independently, by direct computation on the kernel actually used in the main text, is the concrete fact: whatever the exact route through Stokes flow theory, the resulting kernel is, exactly and with the correct constant and sign, -2η times $(-\Delta)^{1/2}$. Equation (4.5) explains why an operator of that specific type, half-order and no other, is the one a linear two-dimensional half-space coupling should be expected to produce.

4.4 Making the analogy precise: the true field is biharmonic, not harmonic

The structural claim above can be sharpened into an explicit computation. Savage's kernel is the response of an actual two-dimensional Stokes flow, not of the abstract harmonic extension (4.5), and the two problems solve different equations. Writing $\zeta := z - H$ for the depth below the crust–mantle interface (main text's z increases downward, with the interface at $z = H$), incompressible Stokes flow in $\zeta > 0$ is governed, through the stream function ψ ($v_x = \partial_\zeta \psi$, $v_z = -\partial_x \psi$, so that incompressibility holds automatically), by the biharmonic equation $\nabla^4 \psi = 0$, not by Laplace's equation.

Solving it directly confirms, and sharpens, the structural analogy. Fourier-transforming in x, the biharmonic equation reduces to the ordinary differential equation $(\partial_\zeta^2 - k^2)^2 \hat{\psi} = 0$, whose solution bounded as $\zeta \to \infty$ is $\hat{\psi}(k, \zeta) = (A + B\zeta) e^{-|k|\zeta}$. Imposing no vertical flow through the interface, $\hat{\psi}(k, 0) = 0$, fixes $A = 0$; imposing the prescribed surface velocity, $\partial_\zeta \hat{\psi}(k, 0) = \hat{V}(k, t)$ where $V := \partial u / \partial t$, fixes $B = \hat{V}(k, t)$:

$$\hat{\psi}(k, \zeta, t) = \hat{V}(k, t)\, \zeta\, e^{-|k|\zeta}. \tag{4.8}$$

This gives the velocity and stress fields at any depth ζ in the asthenosphere, not only at $\zeta = 0$. Taking the shear stress $\tau_{xz} = \eta(\partial_\zeta^2 \psi - \partial_x^2 \psi)$ at the interface reproduces Savage's kernel exactly: $\hat{\tau}_{xz}(k, 0, t) = -2\eta |k| \hat{V}(k, t)$, matching (1.9).

The mismatch with (4.5) is now explicit rather than merely asserted. A pure harmonic extension of the same boundary data would decay as $e^{-|k|\zeta}$ alone; the physical Stokes solution (4.8) carries an extra factor of ζ, the signature of the repeated root at rate $|k|$ in the biharmonic ODE, and the structure Goursat's representation predicts (a harmonic function multiplied by a coordinate, combined with a second harmonic function). So the physical half-space field is not literally the harmonic extension of Section 4; it is a two-harmonic-function, biharmonic object that happens to produce the same boundary operator, $-2\eta(-\Delta)^{1/2}$, once the interface stress is extracted. The half-Laplacian is the correct description of the boundary response either way; the extension variable y of (4.3) should not be read as the physical depth ζ, and depth-resolved fields in the true asthenosphere require solving (4.8) directly, not substituting the surface solution into the Poisson kernel (4.6).

5 Proofs

The short proofs load-bearing for this model's own construction are given inline, at the point of use, in Sections 1–4 (the value of $C(1, \frac{1}{2})$, the P.V./Hadamard sign identity, the Sobolev divergence of the step function, the Hilbert-transform symbol identity, and the $\alpha = 0$ reduction of the extension problem). This section collects the two longer, more general results used along the way, for which we cite rather than reproduce the full argument, following the same practice as Daoud and Laamri (2022) themselves.

Equivalence of the singular-integral and Fourier definitions, general N. That (1.2) and the Fourier-multiplier definition $(-\Delta)^s v = \mathcal{F}^{-1}\{\|k\|^{2s} \hat{v}(k)\}$ agree for every $N \geq 1$ and $s \in (0, 1)$, not only the $N = 1$, $s = \frac{1}{2}$ case proved directly in Section 1, is Definitions 2.1 and 2.2 in Daoud and Laamri (2022), with the full proof referred there to Di Nezza et al. (2012, Section 3). The general argument decomposes the singular integral via a change to polar coordinates and integrates the angular part in closed form against the Fourier representation; it is not reproduced here because it is not needed beyond the $N = 1$ case established directly above.

Full derivation of the extension formula. The extension problem (4.3)–(4.4) for general $s \in (0, 1)$ is proved in Caffarelli and Silvestre (2007), via an explicit Poisson-type kernel for the weighted operator $\mathrm{div}(y^\alpha \nabla \cdot)$ and a direct computation of its Fourier transform in x for fixed y, showing the resulting ODE in y has the modified Bessel function solution whose $y \to 0^+$ weighted normal derivative reproduces $\|k\|^{2s} \hat{v}(k)$ up to the constant d_s. We used only the $\alpha = 0$ specialization, proved directly and elementarily in Section 4 via the classical Poisson-kernel/harmonic-conjugate construction, which does not require the general Bessel-function machinery.

6 Synthesis for structural geology

The mathematics above answers three questions a structural geologist would otherwise have to take on faith from the main text.

Why does deformation appear everywhere at once, with no propagating front? Because the operator is genuinely non-local: Section 1 shows the Stokes kernel is exactly -2η times $(-\Delta)^{1/2}$, an operator whose value at any point x depends on the field at every other point simultaneously, not on a neighborhood that grows with time as in ordinary diffusion. This is the same conclusion the main text reaches in §4.3 by comparing Lorentzian and erfc tails; here it follows directly from the kernel's identity as a fractional Laplacian rather than from the shape of the fitted curves.

Why does an instantaneous dike opening not break the model? Because the half-Laplacian semigroup is smoothing, and quantifiably so: Section 3 shows that no matter how singular the initial displacement field is (a bare Heaviside step fails to belong to the model's own natural energy space, Section 2.1), the field becomes real analytic in space for any $t > 0$ after the intrusion, with a radius of convergence that grows linearly, at least at, at every point. The arctan step response is not a lucky closed form; it is the generic behaviour of any field evolving under this operator. This regularization is intrinsic to the operator, not to the data: the radius at comes entirely from how fast the semigroup multiplier $e^{-a|k|t}$ decays at high frequency, a property that holds for any initial condition in the space of tempered distributions, however singular; the dike-opening step is one instance of that generic behaviour, not something arranged for it in particular. The same section gives a second, complementary picture of why: the correct solution can be built by averaging the ordinary (Fickian) diffusion response to the same initial condition over every possible diffusion timescale, weighted by a fixed subordinator density. That ordinary diffusion response is the Foulger profile the main text rejects as the physical model (§4.3); it survives here only as a mathematical ingredient, blended across timescales rather than taken at face value at any one of them.

Why is the operator order exactly one half, and not some other exponent tuned to fit the data? Because, as shown in Section 4, $s = \frac{1}{2}$ is the unique order at which the extension problem defining $(-\Delta)^s$ reduces to the ordinary harmonic problem in the half-space beneath the surface. This is not only a structural analogy: the same section confirms it by direct computation, solving the actual two-dimensional Stokes flow beneath the plate (governed by the biharmonic equation, not the harmonic one) and showing its interface stress reproduces $-2\eta(-\Delta)^{1/2}$ exactly, with the extension problem's variable y standing in for, but not equal to, the true physical depth. A Newtonian asthenosphere coupled to a thin elastic plate could not have produced any other operator order: half-order nonlocality is the structural signature of this specific rheology and this specific dimensionality, not a fitted parameter.

That last point is also where the mathematics opens a direction for future work rather than closing the subject. The derivation is tied throughout to Newtonian viscosity; a power-law (non-Newtonian) asthenosphere of the kind independently documented in the mantle beneath magmatically active rifts would replace the linear Stokes kernel with a different constitutive relation, and there is no guarantee its Fourier reduction retains the clean $|k|$ dependence exploited here. Whether such a rheology still reduces to a fractional Laplacian of some order $s \neq \frac{1}{2}$, or to an operator outside this family altogether, is an open question this annex does not attempt to answer, but the extension-problem machinery of Section 4 gives a concrete route to it: any proposed kernel could be tested against the same $\alpha = 1 - 2s$ reduction to see whether a consistent order s exists at all.

References

Luis Caffarelli and Luis Silvestre. An extension problem related to the fractional Laplacian. *Communications in Partial Differential Equations*, 32(8):1245–1260, 2007. doi: 10.1080/03605300600987306.

Maha Daoud and El Haj Laamri. Fractional Laplacians: a short survey. *Discrete and Continuous Dynamical Systems - Series S*, 2022. doi: 10.3934/dcdss.2021027. In press.

Eleonora Di Nezza, Giampiero Palatucci, and Enrico Valdinoci. Hitchhiker's guide to the fractional Sobolev spaces. *Bulletin des Sciences Mathématiques*, 136(5):521–573, 2012. doi: 10.1016/j.bulsci.2011.12.004.

R. Metzler and J. Klafter. The random walk's guide to anomalous diffusion. *Physics Reports*, 339: 1–77, 2000.

S. G. Samko, A. A. Kilbas, and O. I. Marichev. *Fractional Integrals and Derivatives: Theory and Applications*. Gordon and Breach, Yverdon, 1993.

J. C. Savage and W. H. Prescott. Asthenosphere readjustment and the earthquake cycle. *Journal of Geophysical Research*, 83:3369–3376, 1978.